Short Form Description

Model	Installation	Filling head(s)	Cylinder type(s)	Centring unit
BD51 021	On a filling **carrousel** with automatic introduction and ejection units	**Semi-automatic** pneumatic self-decoupling filling head that fits one type of centre valve	**Domestic** or **industrial** LPG cylinders with uniform centre valves	No

General Description

The electronic filling machine, type UFM, is an independent unit designed to be installed on a filling carrousel with automatic introduction and ejection units, and is used for semi-automatic filling of LPG cylinders with uniform centre valves. The filling machine is designed to be equipped with a semi-automatic pneumatic self-decoupling filling head that fits one type of centre valve.

The filling machine is available in 2 heights depending on the cylinder type (or cylinder types) to be filled (see also “Cylinder dimensions (requirements and limitations)” and “Dimensions in mm” in “Technical Data”):

- Model UFM-D for filling of LPG cylinders with a total height of min. 200 mm and max. 800 mm
- Model UFM-I for filling of LPG cylinders with a total height of min. 200 mm and max. 1600 mm

The filling head is connected manually to the cylinder valve and the tare values of the cylinders are manually encoded by an operator by means of a keyboard on a control box on each filling machine. When filling cylinders with the same tare value, the operator can encode one fixed tare value on all filling machines, after which the operator only needs to connect the filling head and push a start-button (see also “Dimensions in mm” in “Technical Data”).

The filling machine type UFM consists of the following main components:

- Lower scale with electronic load cell and control/weighing computer for connection to 12 V DC power supply
- Base console with stainless steel weighing plate
- Aluminium column with built-in mounting plate for solenoid valves and pneumatic valves
- Suspension for semi-automatic filling head with balance device
- Necessary compressed air and gas hoses for connection to one manual filling head
- Adjustable control box with keyboard, display and start/stop push-button
- Connection arrangement for LPG and compressed air with corresponding electronically controlled pneumatic gas stop valve and shut-off valve for compressed air
- Safety arrangement which ensures that cylinders with filling heads still connected are not ejected from the filling machine by the automatic ejection unit

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Scope of Supply

The filling machine is supplied complete to be installed on a filling carrousel including corresponding lower scale with electronic load cell and control/weighing computer for connection to 12 V DC power supply, base console with stainless steel weighing plate, aluminium column with built-in mounting plate for solenoid valves and pneumatic valves, suspension for semi-automatic filling head with balance device, necessary compressed air and gas hoses for connection to one manual filling head, adjustable control box with keyboard, display and start/stop push-button, connection arrangement for LPG and compressed air with corresponding electronically controlled pneumatic gas stop valve and shut-off valve for compressed air, and safety arrangement.

Additional Equipment

(Only included if specified in quotation)

- Central power supply (12 V DC):
  - Model A: in flameproof enclosure (Ex d) to be placed in hazardous area at the carrousel system including cabling to the filling machines for supply of max. 24 filling machines
  - Model B: in flameproof enclosure (Ex d) to be placed in hazardous area at the carrousel system including cabling to the filling machines for supply of max. 48 filling machines
- Battery power supply (12 V DC / 6 Ah, EEx m ib IIB T2) to be placed in hazardous area at the carrousel system including cabling to the filling machines. The battery has a life expectancy of approx. 175 hours per charge if it is to supply one single filling machine, or approx. 175/n hours if it is to supply several filling machines (n = the number of filling machines connected to the same battery). I.e. if for example 8 filling machines are connected to the same battery, the battery has a life expectancy of approx. 22 hours (175/8) per charge.
- Battery charger (0.5 A) to be placed in non-hazardous area. The charge time for one battery is approx. 12 hours.
- Communication module and PC including software for data collection
- Filling carrousel with automatic introduction and ejection units
- Semi-automatic pneumatic self-decoupling filling head for centre valves which is adapted to the actual cylinder valve, according to the specifications of the customer
- Standard spare parts set

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Technical Data

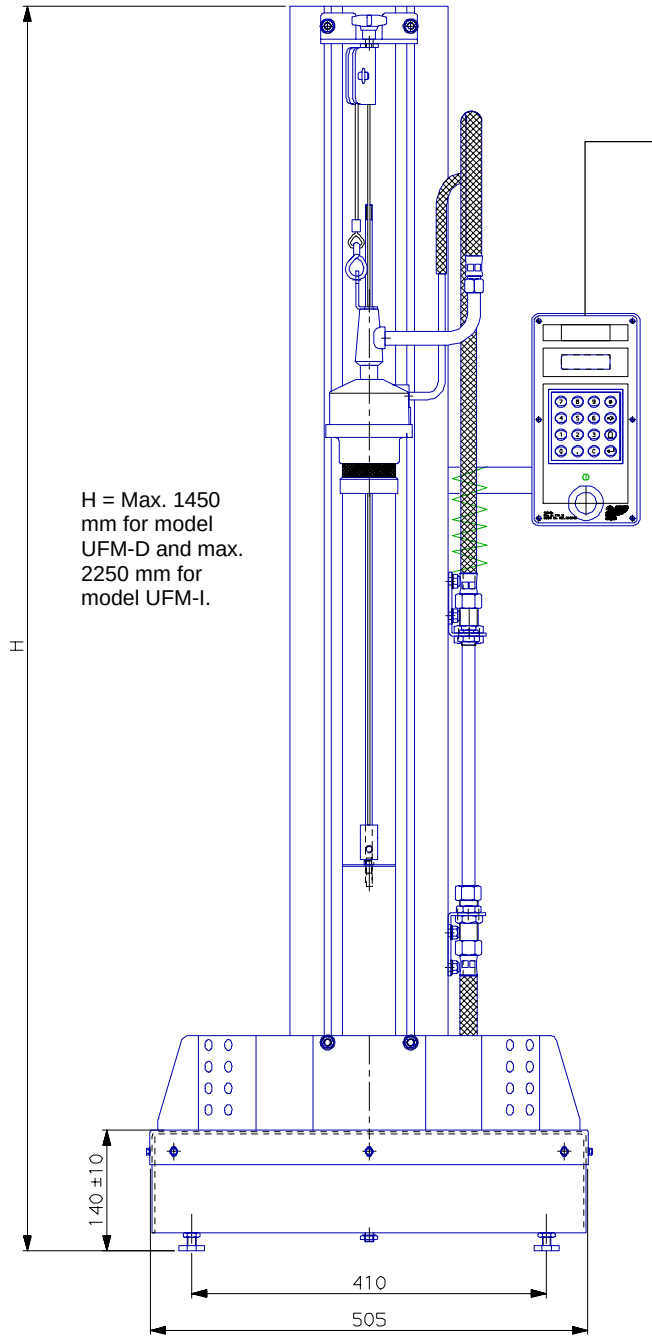
Cylinder dimensions (requirements and limitations)	• Cylinder diameter: – Min. 200 mm - Max. 380 mm – Max. diameter variation for all cylinders: No limitation • Cylinder height: – UFM-D = Min. 200 mm - Max. 800 mm – UFM-I = Min. 200 mm - Max. 1600 mm – Max. height variation for all cylinders within the same cylinder type (the same batch): 300 mm • Shroud diameter: – Inside diameter: Min. 125 mm – Outside diameter: Max. diameter of the cylinder
Necessary connections (specifications and quality requirements)	• LPG installation (via the filling carrousel): – Connection size: G1/2" NPT – Filling pressure: Max. 2.1 MPa. (21 bar) – Test pressure: Max. 3.0 MPa. (30 bar) • Compressed air installation (via the filling carrousel): – Connection size: G1/4" BSP – Working pressure: Min. 0.6 MPa. (6 bar) - Max. 1.0 MPa. (10 bar) – Quality: The quality of the compressed air should be in accordance with ISO 8573.1, class 2-3-2 [1)] • Electrical power installation (via the filling carrousel and the power supply): – Voltage: 1-phase + neutral + earth (voltage variation from 85 to 264 V AC) – Frequency: Variation from 48 to 62 Hz • Earth net (via the filling carrousel): – Max. transition resistance between earth wire and earth: ≤ 2.0 Ohm
Consumption data	• Compressed air [2)]: – Approx. 2.1 normal litres/cylinder (including semi-automatic pneumatic self-decoupling filling head for centre valves) • Electrical power: – < 1.0 w
Weight	• Approx. 90 kg
Approvals	• The filling machine is designed for use in hazardous areas classified as Zone 1 according to IEC 79-10 and Class I, Division 1 according to NEC (USA), article 500 • The weighing accuracy of the load cell is according to OIML R 60, NTEP 3000d, accuracy class C3
Other data	• The weighing computer calculates and shows the weight in kilograms according to the SI standard • Weighing range: Min. 5 kg - Max. 120 kg Note: The scale requires a minimum load of 5 kg to start • Display division: 100 g (e.g. 15.1 kg, 15.2 kg, 15.3, etc.) • The filling machine is designed for installation in locations in which the temperature may vary between -10°C and +50°C

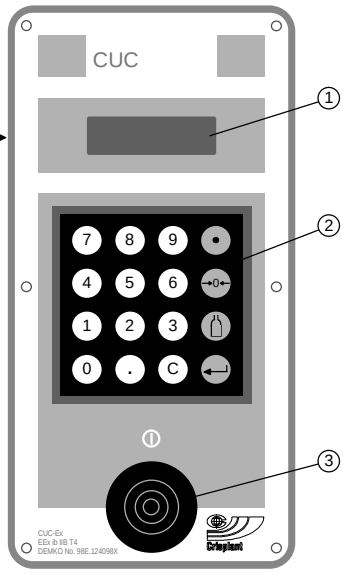
[1)] The compressed air must not contain particles larger than 1 µm. The pressure dew point of the compressed air at a pressure of 0.7 MPa (7 bar) shall be at least 10°C lower than the lowest ambient temperature to which the compressed air system is exposed at the coldest place of the plant during the year (class 3 ~ -20°C). The maximum oil contents in the compressed air must not exceed 0.1 mg/m³.

[2)] The compressed air consumption can be used for dimensioning the compressor capacity (FAD = Free Air Delivery) based on local conditions. The specified compressed air consumption is based on the following standard reference conditions: An inlet pressure of 0.1 MPa (1 bar), a suction temperature of 20°C and a relative humidity of 60%.

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Dimensions in mm





The tare value of the cylinder is manually encoded by the operator by means of the keyboard (pos. 2) on the control box on each filling machine.

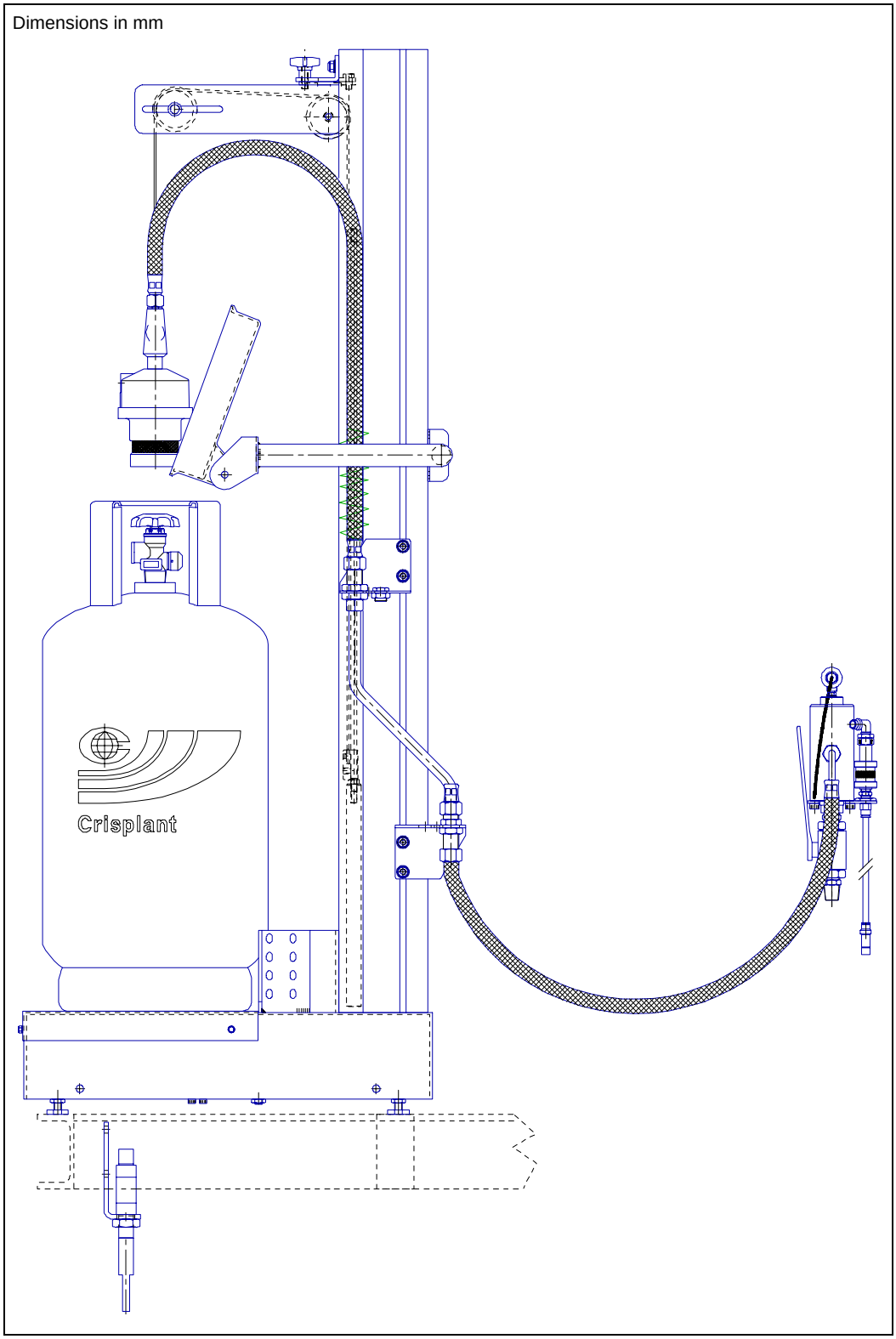
When filling cylinders with the same tare value, the operator can encode one fixed tare value on all filling machines on the filling carrousel, after which the operator only needs to connect the filling head and push the start-button (pos. 3).

All data (encoded values, process status, filling results, help texts, error messages, etc.) is shown in the display (pos. 1).

The start/stop push-button (pos. 3) may be used to either stop the filling process or restart the filling process after a stoppage.

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Dimensions in mm



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Other Models Available

Model	Installation	Filling head(s)	Cylinder type(s)	Centring unit
BD51 001	On a filling **carrousel** with automatic introduction and ejection units	**Automatic** filling head that fits one type of **centre** valve	Uniform **domestic** LPG cylinders with uniform centre valves	Yes
BD51 011	On a filling **carrousel** with automatic introduction and ejection units	**Automatic** filling head that fits one type of **centre** valve	Uniform **domestic** LPG cylinders with uniform centre valves	No
BD51 021	On a filling **carrousel** with automatic introduction and ejection units	**Semi-automatic** pneumatic self-decoupling filling head that fits one type of centre valve	**Domestic** or **industrial** LPG cylinders with uniform centre valves	No
BD51 031	On a filling **carrousel** with automatic introduction and ejection units	**Manual** pneumatic filling head that fits one type of **screw** valve or **manual** mechanical filling head that fits one type of **centre** valve	**Domestic** or **industrial** LPG cylinders with uniform screw or centre valves	No
BD51 032	On a filling **carrousel** with automatic introduction and ejection units	**Manual** pneumatic filling head that fits one type of **screw** valve or **manual** mechanical filling head that fits one type of **centre** valve	**Domestic** or **industrial** LPG cylinders with uniform screw or centre valves	Yes
BD51 041	For **stationary** installation on floor	**Manual** pneumatic filling head that fits one type of **screw** valve or **manual** mechanical filling head that fits one type of **centre** valve	**Domestic** or **industrial** LPG cylinders with uniform screw or centre valves	No
BD51 051	For **stationary** installation on floor	**Semi-automatic** pneumatic self-decoupling filling head that fits one type of centre valve	**Domestic** or **industrial** LPG cylinders with uniform centre valves	No
BD51 061	To be incorporated in-line in a **chain conveyor** plant	**Manual** pneumatic filling head that fits one type of **screw** valve or **manual** mechanical filling head that fits one type of **centre** valve	**Domestic** or **industrial** LPG cylinders with uniform screw or centre valves	No
BD51 071	To be incorporated in-line in a **chain conveyor** plant	**Semi-automatic** pneumatic self-decoupling filling head that fits one type of centre valve	**Domestic** or **industrial** LPG cylinders with uniform centre valves	No
BD51 081	To be incorporated in-line in a **roller conveyor**	**Manual** pneumatic filling head that fits one type of **screw** valve or **manual** mechanical filling head that fits one type of **centre** valve	**Domestic** or **industrial** LPG cylinders with uniform screw or centre valves	No
BD51 091	To be incorporated in-line in a **roller conveyor**	**Semi-automatic** pneumatic self-decoupling filling head that fits one type of centre valve	**Domestic** or **industrial** LPG cylinders with uniform centre valves	No

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